

DATA SCIENCE INTERNSHIP WEEK 1 FINAL REPORT

Data Cleaning, Sales Analysis and Visualization

Organization: WeIntern Pvt Ltd

Assignment: Data Science Internship - Week 1

Tools: Python, Pandas, Jupyter Notebook, Power BI

This report consolidates the three Week 1 tasks into one evidence-based analytical submission.

1. Title and Objective

The Week 1 assignment develops disciplined habits in data cleaning, structured analysis, business interpretation, visualization, and professional reporting. The work is divided into three connected tasks: cleaning an e-commerce dataset, analyzing a sales dataset, and communicating the resulting sales insights through a Power BI dashboard.

The report documents what was cleaned, what was analyzed, what was discovered, why the findings matter, and what actions may reasonably be considered next. Observations are kept separate from recommendations, and causal explanations are not claimed where the available data does not support them.

2. Dataset Overview

2.1 Task 1 - Customer Shopping Behavior Dataset

Task 1 uses a customer shopping behavior dataset containing demographic, product, purchase, rating, subscription, discount, shipping, payment, and purchase-frequency information. The raw file contains 3,900 rows and 18 columns.

Dataset	Rows	Columns	Purpose
Customer Shopping Behavior - Raw	3,900	18	Task 1 cleaning and validation
Customer Shopping Behavior - Cleaned	3,900	19	Cleaned Task 1 output
Sample Superstore	9,994	21	Tasks 2 and 3 sales analysis/visualization

2.2 Tasks 2 and 3 - Sample Superstore Dataset

Tasks 2 and 3 use the Sample Superstore dataset. It contains 9,994 order-line records with order, customer, product, geographic, sales, quantity, discount, profit, and shipping fields. The dataset contains 5,009 unique orders, 793 unique customers, and 1,850 unique products.

Because one order may contain multiple product lines, Order ID can legitimately repeat. Order-based metrics therefore use distinct Order ID counts rather than the number of rows.

3. Cleaning Summary

Task 1 began with inspection of structure, data types, summary statistics, missing values, duplicates, and logical validity. The only missing field was Review Rating, with 37 missing values (0.95% of 3,900 records).

- Missing Review Rating values were imputed using the median rating within the corresponding product category.
- Column names were standardized to lowercase snake_case for consistent programmatic use.

- Two derived fields were created: age_group and purchase_frequency_days.
- Age, purchase amount, and previous-purchase values were checked for invalid ranges; no invalid records were found.
- Categorical fields were checked for leading/trailing whitespace; no issues were found.
- discount_applied and promo_code_used were confirmed to contain identical information, so promo_code_used was removed as redundant.
- Fully duplicated rows were checked and none were found.
- The raw dataset was preserved and the cleaned dataset was saved separately.

The cleaned output contains 3,900 rows and 19 columns, with 0 missing values and 0 fully duplicated rows.

4. KPI Summary

The sales analysis uses clear order-level logic. Total Orders is the number of unique Order IDs, and Average Order Value is Total Revenue divided by Total Orders.

KPI	Result
Total Revenue	\$2,297,200.86
Total Orders	5,009
Average Order Value	\$458.61
Total Units Sold	37,873
Top-Selling Category	Technology (\$836,154.03)
Top Revenue-Generating Product	Canon imageCLASS 2200 Advanced Copier (\$61,599.82)

These KPIs provide the baseline for the trend, product, regional, and customer analyses that follow.

5. Trend Analysis

Year	Revenue	Orders	Units	Revenue Growth
2014	\$484,247.50	969	7,581	-
2015	\$470,532.51	1,038	7,979	-2.83%
2016	\$609,205.60	1,315	9,837	29.47%
2017	\$733,215.26	1,687	12,476	20.36%

Revenue declined slightly in 2015 (approximately 2.83%) and then increased strongly in 2016 (approximately 29.47%) and 2017 (approximately 20.36%). Unique orders and units sold also rose during the later years, indicating stronger sales activity across multiple measures.

November 2017 recorded the highest monthly revenue at approximately \$118,447.83 from 261 unique orders and 1,840 units. February 2014 recorded the lowest monthly revenue at approximately \$4,519.89. The dataset shows the pattern but does not establish the specific causes of the peaks or declines.

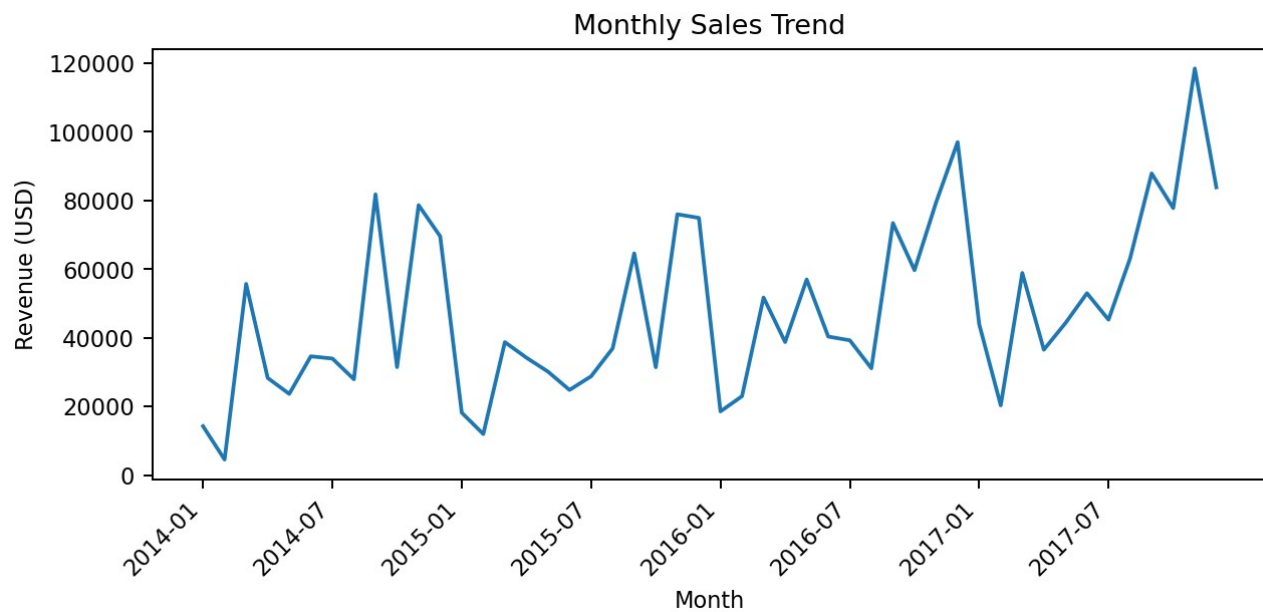


Figure 1. Monthly sales trend based on the Sample Superstore dataset.

6. Customer Insights

The sales dataset contains 793 unique customers. Customer behavior was evaluated using unique order count, revenue contribution, repeat purchasing, and segment performance.

- Sean Miller generated the highest customer revenue at approximately \$25,043.05 from five unique orders.
- Emily Phan placed the most unique orders (17), generating approximately \$5,478.06 in revenue.
- 781 of 793 customers (98.49%) placed more than one unique order.
- Repeat customers generated approximately 99.78% of total observed revenue.
- The Consumer segment generated approximately 50.56% of total revenue and recorded the highest number of unique orders.

The highest-revenue customer was not the most frequent customer, showing that purchase frequency and monetary value capture different dimensions of customer importance.

7. Visualization Section

Task 3 communicates the Task 2 results through a one-page Power BI dashboard. The dashboard contains four KPI cards, category and product comparisons, a monthly trend, customer-segment revenue share, regional performance, and slicers for Year, Region, Category, and Segment.

7.1 Revenue by Category - Bar Chart

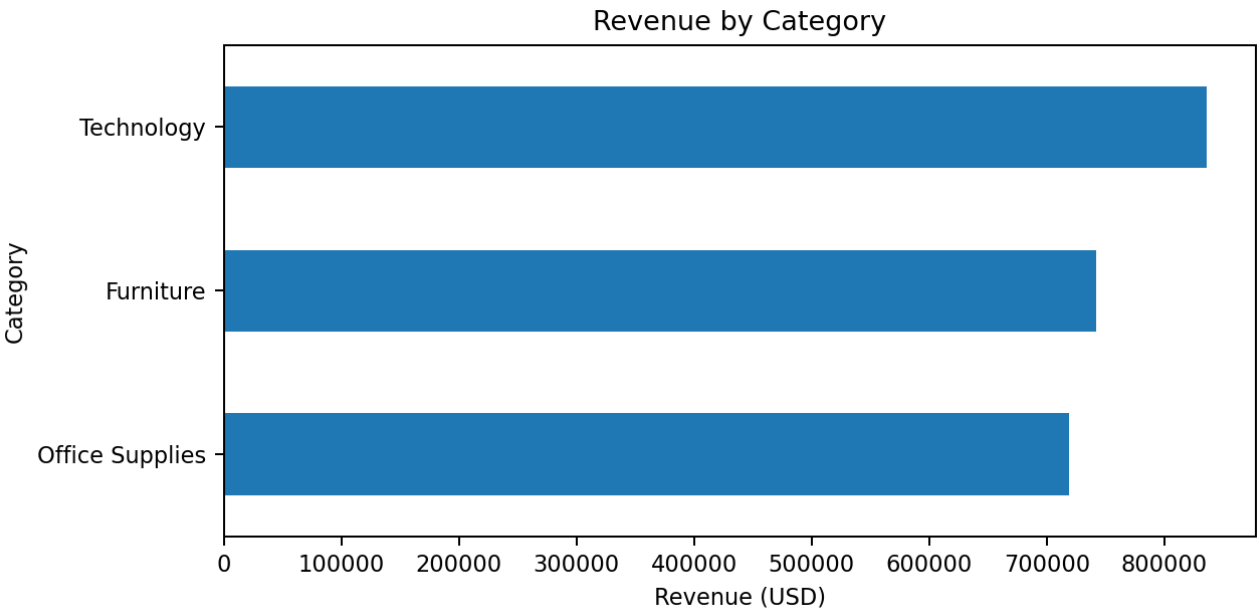


Figure 2. Revenue by category.

Technology generated the highest category revenue at approximately \$836,154 (36.40% of total revenue). Furniture and Office Supplies also made substantial contributions, so revenue was not overwhelmingly concentrated in one category.

7.2 Revenue Share by Customer Segment - Pie Chart

Revenue Share by Customer Segment

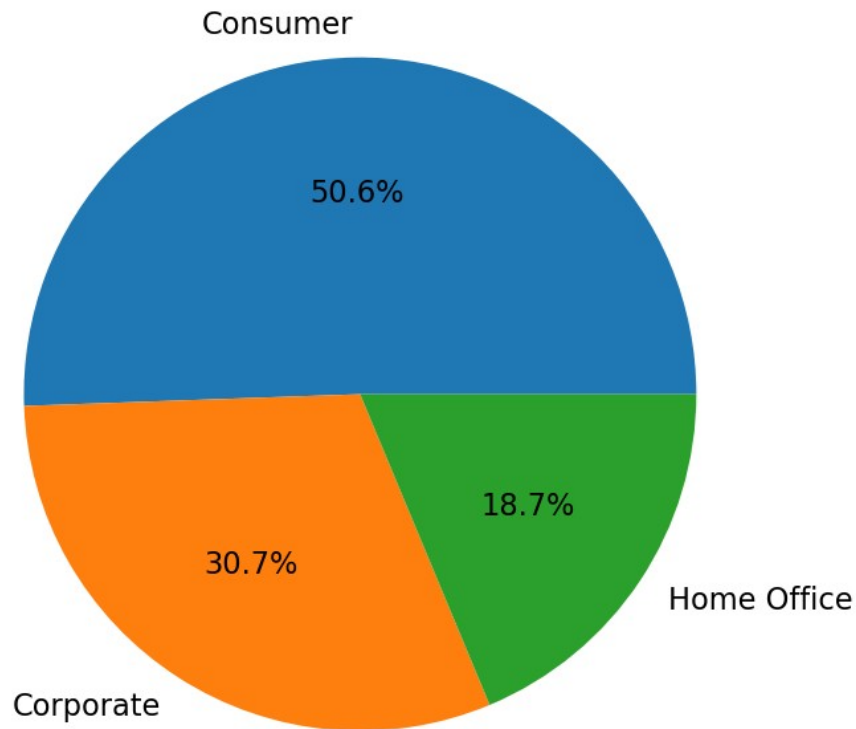


Figure 3. Revenue share by customer segment.

Consumer contributed approximately 50.56% of total revenue, followed by Corporate at 30.74% and Home Office at 18.70%. With only three segments, the proportional comparison remains readable.

7.3 Top Products by Units Sold

Staples ranked first by unit volume with 215 units sold. In contrast, the Canon imageCLASS 2200 Advanced Copier generated the highest product revenue. This demonstrates why unit demand and revenue contribution should be evaluated separately.

7.4 Revenue by Region

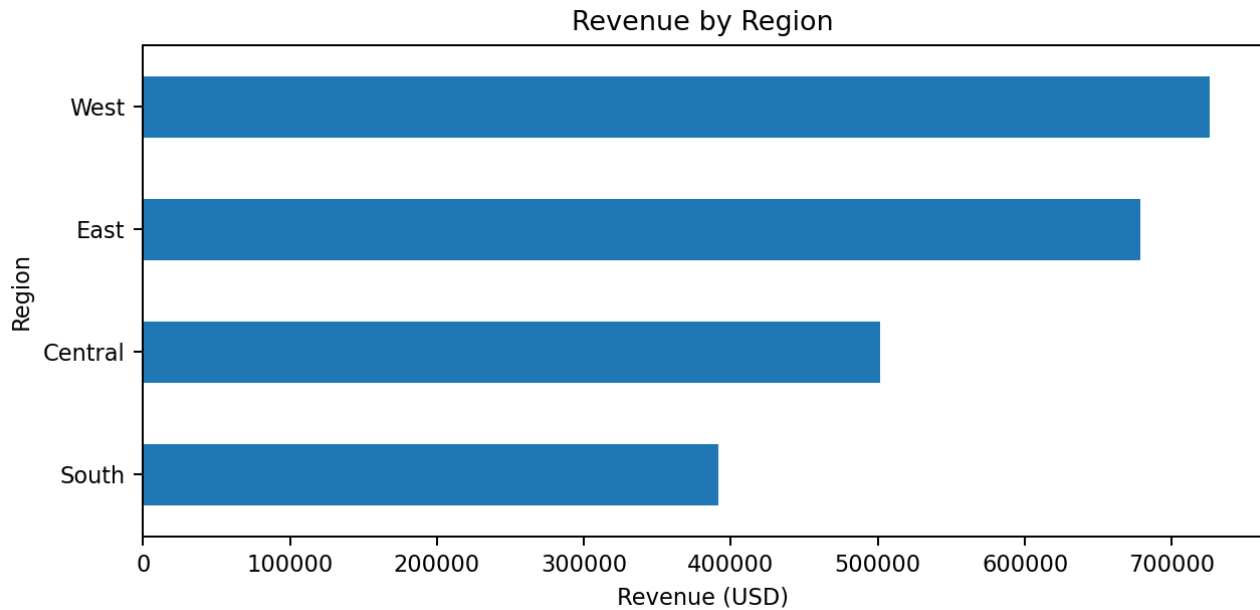


Figure 4. Revenue by region.

West generated the highest regional revenue at approximately \$725,458 (31.58%), followed by East at 29.55%. West and East together accounted for the majority of observed sales.

7.5 Dashboard Interpretation

The dashboard combines overall sales scale, changes over time, category contribution, customer-segment composition, regional differences, and high-volume products on one page. The visuals complement rather than duplicate one another and support the analytical story developed in Task 2.

8. Key Findings

1. Overall sales performance: approximately \$2.30 million in revenue from 5,009 unique orders and 37,873 units sold, with an average order value of \$458.61.
2. Sales strengthened during the later years: revenue declined by approximately 2.83% in 2015, then increased by approximately 29.47% in 2016 and 20.36% in 2017.
3. November 2017 was the highest-revenue month, generating approximately \$118,448 from 261 unique orders and 1,840 units.
4. Technology was the highest-revenue category at approximately \$836,154, representing 36.40% of total revenue.
5. Product revenue and volume leadership differed: Canon imageCLASS 2200 Advanced Copier led revenue at approximately \$61,600, while Staples led volume at 215 units.
6. West generated the highest regional revenue at approximately \$725,458, or 31.58% of total revenue.

7. Consumer was the largest customer segment, generating approximately 50.56% of total revenue and the greatest number of unique orders.
8. Repeat customers dominated observed activity: 781 of 793 customers (98.49%) placed more than one unique order and generated approximately 99.78% of revenue.

9. Recommendations

9. Prioritize repeat-customer retention by evaluating loyalty, retention, and repeat-purchase initiatives while continuing to monitor new-customer acquisition.
10. Investigate the drivers of stronger 2016-2017 performance, especially November 2017, using available promotion, product-mix, customer-activity, and operational information before attributing the increase to any specific cause.
11. Evaluate product performance using both revenue and unit volume because the highest-revenue product and highest-volume product are different.
12. Maintain focus on Technology while preserving category diversification because Furniture and Office Supplies also contribute substantial revenue.
13. Investigate regional differences further using profitability, customer density, product mix, and order characteristics to determine whether additional regional opportunities exist.
14. Consider customer-segment differences in merchandising and engagement planning while recognizing that Corporate and Home Office remain meaningful contributors.

10. Limitations and Assumptions

- The Task 1 customer-shopping dataset and the Task 2/3 Superstore dataset are separate datasets and should not be interpreted as records from the same customers or transactions.
- The Superstore dataset is stored at the order-line level; distinct Order ID is therefore used for order-based metrics.
- Revenue is calculated from the provided Sales field.
- Repeat customers are defined as customers with more than one unique order in the supplied dataset.
- The analysis describes patterns in the available data and does not establish causal explanations for changes in sales.
- Monthly and yearly differences may reflect factors that are not represented in the supplied variables.
- The conclusions are limited to the time period, customers, products, and geographic coverage represented in the supplied datasets.